PCBC H Parametrics Practice Test C Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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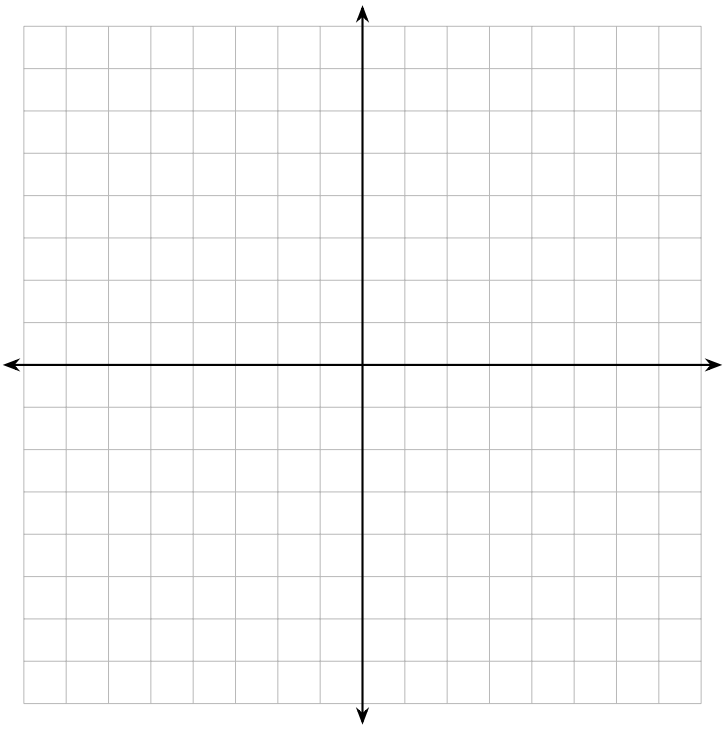
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SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

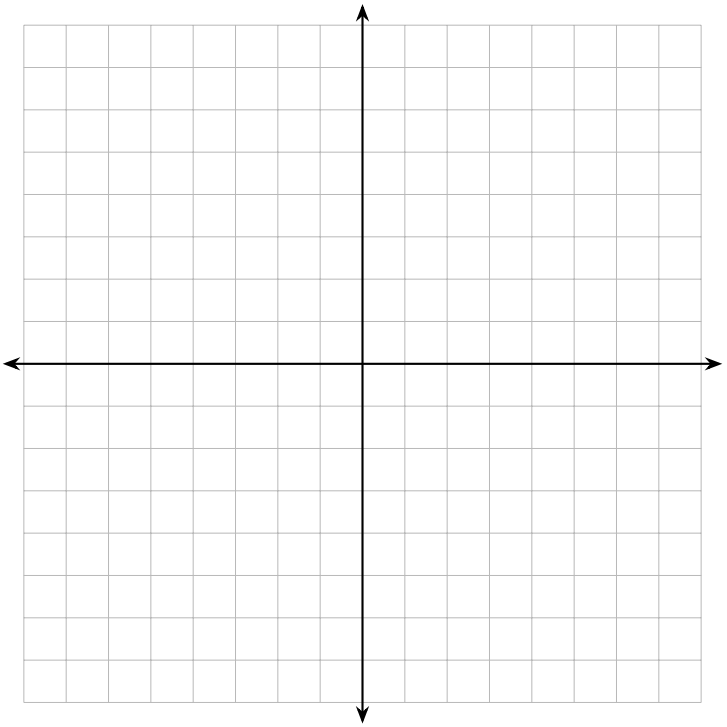
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = 3 - t, \quad y = t^2 - 4t, \quad \text{for } t \text{ in } (0,4]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 1 + 2\sin t, \quad y = -2 + 3\cos t, \quad \text{for } t \text{ in } [0, 2\pi]$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points) Then answer part (b).

3. (a) $y = x^2 - 6x + 7$ **3 pts**

Representation 1 *Representation 2*

(b) The curve $x = f(t),\ y = g(t)$ is changed to $x = 5 + f(t),\ y = g(t)$. How does its graph change?

Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

(a) A ball is thrown from a height of 6 ft at an angle of inclination of 45° with a velocity of 48 ft/sec. Write the parametric equations, then eliminate t to write y as a function of x. Use that equation to find the horizontal distance the ball travels.

(b) Graph the cycloid $x = 3t - 3\sin t,\ y = 3 - 3\cos t$ for t in $[0, 4\pi]$ on your calculator. Give a window that shows the whole curve, its maximum height, and every x where it touches the x-axis. **3 pts**

PCBC H Parametrics Practice Test C: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right. Final answers are written the way the homework answer keys write them: solved for y when possible, hyperbola branches as $y = \sqrt{\cdot}$, ellipses in "$= 1$" form.

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1. $x = 3 - t,\ y = t^2 - 4t,$ for t in $(0, 4]$ *Non-Calculator*

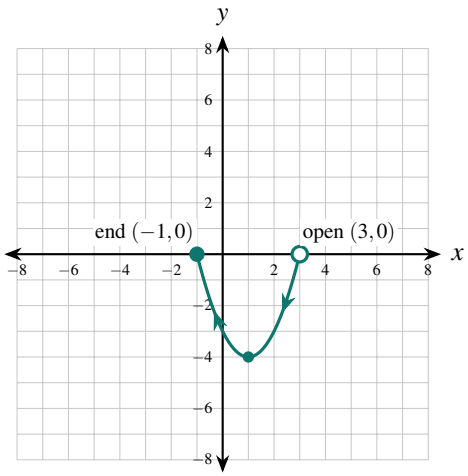
$\boxed{y = x^2 - 2x - 3, \quad -1 \le x < 3}$

From an open circle at $(3, 0)$ down to the vertex $(1, -4)$, then up to a solid dot at $(-1, 0)$: right to left.

Work	Why
$t = 3 - x$	x gives t in one step.
$y = (3-x)^2 - 4(3-x) = x^2 - 2x - 3$	Substitute and expand.
$t \to 0$: $(3, 0)$ excluded	$(0, 4]$ leaves out $t = 0$; no other t gives $x = 3$, so it is a hole: open circle.
$t = 4$: $(-1, 0)$ included	Closed end: solid dot.
$-1 \le x < 3$	$x = 3 - t$ only decreases.
$t = 2$: $(1, -4)$	y turns there; $-4 \le y \le 0$.
arrows right to left	x gets smaller as t increases.

t	(x, y)
0*	$(3, 0)$ open
1	$(2, -3)$
2	$(1, -4)$
3	$(0, -3)$
4	$(-1, 0)$

*Not in the interval; used only to find where the hole goes.



Practice scoring (3 pts): 1 for the equation, 1 for $-1 \le x < 3$, 1 for the graph with the open circle, solid dot and right-to-left arrows.

Trap: a solid dot at $(3, 0)$. The interval is open at $t = 0$ and no other t reaches that point.

2. $x = 1 + 2\sin t,\ y = -2 + 3\cos t,$ for t in $[0, 2\pi]$ *Non-Calculator*

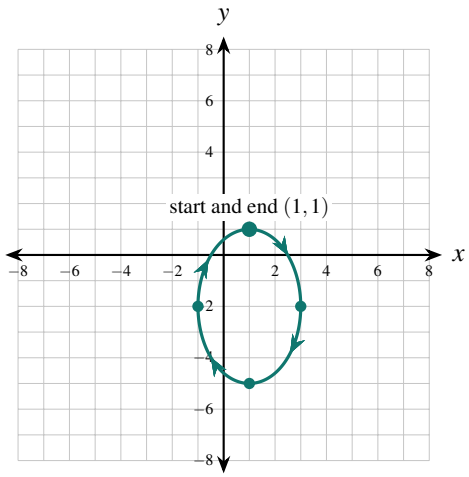
$$\frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} = 1$$

Whole ellipse, center $(1,-2)$, traced once clockwise, starting and ending at $(1,1)$.

Work	Why
$\sin t = \dfrac{x-1}{2}, \quad \cos t = \dfrac{y+2}{3}$	Isolate each trig function.
$\sin^2 t + \cos^2 t = 1$	Pythagorean identity.
$\dfrac{(x-1)^2}{4} + \dfrac{(y+2)^2}{9} = 1$	Ellipse, center $(1,-2)$, 2 across, 3 up and down.
$t = 0 : (1,1);\ \frac{\pi}{2} : (3,-2);\ \pi : (1,-5)$	Top, then right, then bottom: clockwise.
$t = 2\pi : (1,1)$	Back to the start: one start/end point.

t	(x,y)
0	$(1,1)$
$\frac{\pi}{2}$	$(3,-2)$
π	$(1,-5)$
$\frac{3\pi}{2}$	$(-1,-2)$
2π	$(1,1)$

Closed curve: $t = 0$ and $t = 2\pi$ land on the same point, so one dot, not two.



Practice scoring (3 pts): 1 for the equation, 1 for the ellipse with center $(1,-2)$ drawn to size, 1 for clockwise arrows from $(1,1)$.

Trap: assuming counterclockwise. With $\sin t$ in x and $\cos t$ in y, the start is the top and the motion is clockwise.

3. (a) $y = x^2 - 6x + 7$ (b) $x = 5 + f(t)$ *Non-Calculator*

(a) $\boxed{x = t,\ y = t^2 - 6t + 7,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t + 3,\ y = t^2 - 2,\ \text{for } t \text{ in } (-\infty, \infty)}$

(b) $\boxed{\text{The graph shifts 5 units to the right.}}$

Work	**Why**
Representation 1: $x = t,\ y = t^2 - 6t + 7$	Let x be the parameter; every real t.
Representation 2: $y = (x^2 - 6x + 9) - 9 + 7 = (x - 3)^2 - 2$	Complete the square.
$t = x - 3 \Rightarrow x = t + 3,\ y = t^2 - 2$	Let t be the inside.
Check: $(t + 3)^2 - 6(t + 3) + 7 = t^2 - 2$	Put $x = t + 3$ back in.
(b) every point $(f(t), g(t))$ moves to $(5 + f(t), g(t))$	Adding 5 to x moves each point 5 right; y is unchanged.

Practice scoring (3 pts): 1 for each correct pair (with the t interval), 1 for "shifted 5 units right".

Trap: saying "left". Adding to x moves the graph right.

4. *Calculator*

(a) $\boxed{x = (48\cos 45^\circ)t,\ \ y = -16t^2 + (48\sin 45^\circ)t + 6}$, $\boxed{y = -\dfrac{x^2}{72} + x + 6}$, distance $\boxed{\approx 77.569 \text{ ft}}$

(b) window x: 0 to 40, y: -1 to 8; maximum height $\boxed{6}$; touches the x-axis at $\boxed{x = 0,\ 6\pi \approx 18.850,\ 12\pi \approx 37.699}$

Work	**Why**
(a) $x = (48\cos 45^\circ)t = 24\sqrt{2}t \approx 33.941t$	$\cos 45^\circ = \frac{\sqrt{2}}{2}$.
$y = -16t^2 + 24\sqrt{2}t + 6$	$\sin 45^\circ = \frac{\sqrt{2}}{2}$ too.
$t = \dfrac{x}{24\sqrt{2}}$	Solve x for t ($v_0 \cos\theta \neq 0$).
$y = -16 \cdot \dfrac{x^2}{1152} + x + 6 = -\dfrac{x^2}{72} + x + 6$	Substitute; $(24\sqrt{2})^2 = 1152$ and $24\sqrt{2} \cdot \frac{1}{24\sqrt{2}} = 1$.
$0 = -\dfrac{x^2}{72} + x + 6 \Rightarrow x^2 - 72x - 432 = 0$	Lands when $y = 0$; multiply by -72.
$x = 36 \pm 24\sqrt{3}$, so $x \approx 77.569$	Quadratic formula; reject the negative root.
(b) arches repeat every 2π in t: $x = 0,\ 6\pi,\ 12\pi$	$y = 0$ when $\cos t = 1$: $t = 0, 2\pi, 4\pi$, $x = 3t$ there.
top: $y = 3 - 3(-1) = 6$ at $t = \pi, 3\pi$	$\cos t = -1$; tops at $x = 3\pi, 9\pi$.
radian mode, $0 \le t \le 4\pi$	Trig in t needs radians; two arches.

Calculator setup and check

(a) Check with t: $0 = -16t^2 + 24\sqrt{2}t + 6$ gives $t \approx 2.285$ and $x = 24\sqrt{2}(2.28541\ldots) \approx 77.569$, the same. (b) Graphs page, Parametric: $x1(t) = 3t - 3\sin(t)$, $y1(t) = 3 - 3\cos(t)$, $0 \le t \le 4\pi$; trace to read the tops and the x-intercepts.

Practice scoring (3 pts): 1 for the equations and the rectangular form, 1 for the distance, 1 for the cycloid window, height and x-intercepts.

Trap: degree mode for the cycloid, or forgetting that the x-intercepts are 6π apart, not 2π.